Supplementary Material: Patient-Independent TinyML ECG Classification — Per-Fold Support, Noise, and Calibration Details

HeartLens — Supplement to Main Paper (6 pages)

This supplement provides the per-fold numbers referenced in the main paper (`results/folds_5x2.json`, frozen 5×2, Seeds 0,1). It is *not* counted in the 6-page limit. Main Figs. 4–5 and Tables II–VI stay in the main paper; Figs. S1–S4 below were deferred to meet the limit.

I. Frozen Patient-Disjoint Folds

48 MIT-BIH records, $\texttt{Patients}_{\text{train}} \cap \texttt{Patients}_{\text{test}} = \emptyset$. Each of 10 test folds is listed with its record IDs and test support (N/APB/V windows, 1-s centered on R-peaks, 360 samples). APB support 1–502/fold explains the high variance in Table II in the main paper; two folds ≤2 APB windows make per-fold APB F1 undefined/0 (mean-of-folds unstable). Pooled-confusion macro 0.643 (both CNN/TCN) is reported alongside mean-of-folds 0.619/0.615 in the main text.

TABLE I
Frozen folds
(heart-lens-training/results/folds_5x2.json)

Seed	Fold	Test records	N / APB / V (windows)
0	0	101,102,104,112,119	587 / 2 / 232
0	1	103,106,111,117,124	466 / 138 / 304
0	2	100,108,118,200,201	672 / 502 / 602
0	3	107,109,113,114,203	473 / 32 / 362
0	4	105,115,116,208,210	802 / 38 / 769
1	0	103,106,116,200,202	775 / 152 / 502
1	1	100,107,109,117,122	631 / 25 / 458
1	2	101,112,118,121,123	659 / 482 / 426
1	3	102,104,113,115,203	380 / 1 / 566
1	4	105,108,111,114,119	555 / 52 / 317

II. Per-Fold F1 and Support (Grouped CV)

All four architectures share identical folds (paired). Raw per-fold JSON: `heart-lens-training/results/group_kfold_ckpt/{cnn,tcn,lstm,gru}_s{seed}_f{fold}.json`. Below: support (windows) and F1 (Normal/APB/PVC) + macro.

III. Deferred Figures (single-column)

Figs. S1–S4 were in prior drafts as double-column; they are retained here single-column for completeness. Main paper keeps Figs. 4–5 in the main paper and calibration as primary deployment figures.

TABLE II
CNN — per-fold F1 and support (macro mean 0.619±0.242)

Seed	Fold	Support N/APB/V	F1 N / APB / PVC	Macro
0	0	587/2/232	0.977/0.000/0.956	0.644
0	1	466/138/304	0.877/0.837/0.907	0.874
0	2	672/502/602	0.549/0.000/0.000	0.183
0	3	473/32/362	0.866/0.596/0.865	0.776
0	4	802/38/769	0.958/0.000/0.972	0.643
1	0	775/152/502	0.969/0.866/0.986	0.940
1	1	631/25/458	0.930/0.000/0.936	0.622
1	2	659/482/426	0.592/0.000/0.000	0.197
1	3	380/1/566	0.799/0.000/0.841	0.547
1	4	555/52/317	0.890/0.535/0.866	0.764

TABLE III
TCN — per-fold F1 and support (macro mean 0.615±0.229)

Seed	Fold	Support N/APB/V	F1 N / APB / PVC	Macro
0	0	587/2/232	0.988/0.000/0.974	0.654
0	1	466/138/304	0.788/0.732/0.851	0.790
0	2	672/502/602	0.551/0.000/0.026	0.192
0	3	473/32/362	0.886/0.474/0.914	0.758
0	4	802/38/769	0.961/0.000/0.970	0.644
1	0	775/152/502	0.970/0.864/0.983	0.939
1	1	631/25/458	0.922/0.000/0.916	0.612
1	2	659/482/426	0.592/0.000/0.000	0.197
1	3	380/1/566	0.877/0.100/0.935	0.637
1	4	555/52/317	0.902/0.400/0.890	0.730

IV. Noise Per-Artifact Full Table

`heart-lens-training/results/per_artifact_noise.js` macro F1 vs SNR per artifact (Butterworth wins 3/5, raw wins motion, AE only low-SNR PLI).

V. Calibration and Quantization Details

Main Table VII in the main paper; same 918-window test split (493/24/401 class balance). FP32 *vs* INT8 (PTQ, 200 train windows, `TFLITE_BUILTINS_INT8`) ECE/NLL/Brier with separate temperatures (T=0.350, T_{INT8}=5.0). QAT CNN (5×2, TF-MOT, 3 epochs, `results/qat_cnn_summary.json`): FP32 0.642, PTQ 0.549, QAT 0.584 ($\Delta_{\text{QAT}-\text{PTQ}}$+0.036 ± 0.085 p=0.42, 38% of PTQ loss, still −0.058 vs FP32). QAT for Conv1D required TF-MOT default 8-bit path; standard PTQ is the deployed pipeline.

TABLE IV
LSTM — PER-FOLD F1 AND SUPPORT (MACRO MEAN 0.582±0.096)

Seed	Fold	Support N/APB/V	F1 N / APB / PVC	Macro
0	0	587/2/232	0.955/0.000/0.895	0.617
0	1	466/138/304	0.694/0.567/0.675	0.645
0	2	672/502/602	0.659/0.008/0.809	0.492
0	3	473/32/362	0.688/0.161/0.696	0.515
0	4	802/38/769	0.944/0.048/0.960	0.650
1	0	775/152/502	0.886/0.599/0.899	0.795
1	1	631/25/458	0.784/0.035/0.697	0.505
1	2	659/482/426	0.659/0.000/0.689	0.449
1	3	380/1/566	0.798/0.000/0.854	0.551
1	4	555/52/317	0.848/0.164/0.784	0.598

TABLE V
GRU — PER-FOLD F1 AND SUPPORT (MACRO MEAN 0.526±0.170)

Seed	Fold	Support N/APB/V	F1 N / APB / PVC	Macro
0	0	587/2/232	0.973/0.000/0.921	0.631
0	1	466/138/304	0.468/0.397/0.264	0.376
0	2	672/502/602	0.657/0.008/0.866	0.511
0	3	473/32/362	0.447/0.088/0.173	0.236
0	4	802/38/769	0.934/0.000/0.952	0.629
1	0	775/152/502	0.914/0.598/0.926	0.813
1	1	631/25/458	0.840/0.000/0.803	0.548
1	2	659/482/426	0.393/0.016/0.466	0.292
1	3	380/1/566	0.769/0.000/0.858	0.542
1	4	555/52/317	0.888/0.289/0.864	0.680

VI. REPRODUCIBILITY

Frozen folds `results/folds_5x2.json`, 40+16+40 ckpts in `results/group_kfold_ckpt/` and `results/qat_ckpt/` (`.tmp`→`.json` atomic), `results/calibration_int8.json`, seeds 0,1, TF 2.21/Keras 3/sklearn. All main tables/figs regenerate via `run_experiments.sh` and `scripts/round3_*.py`.

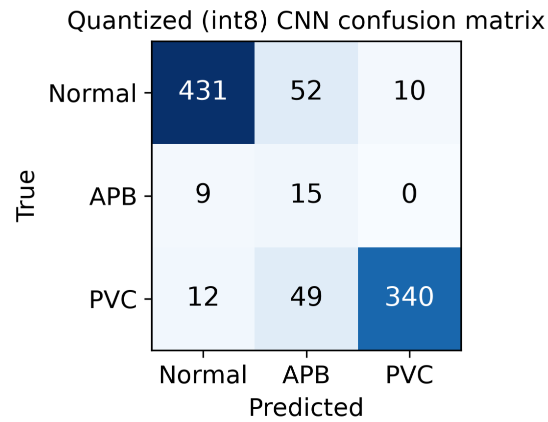

Fig. 1. Confusion matrix (representative fold; supplementary).

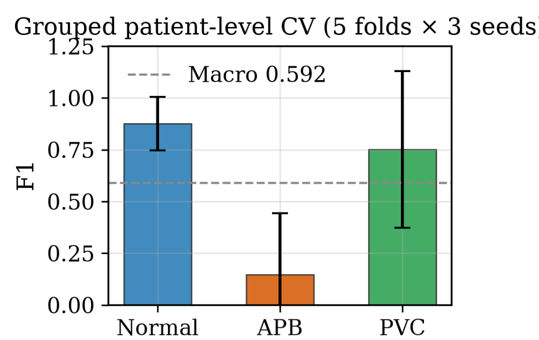

Fig. 2. Grouped-CV macro distribution per architecture (supplementary).

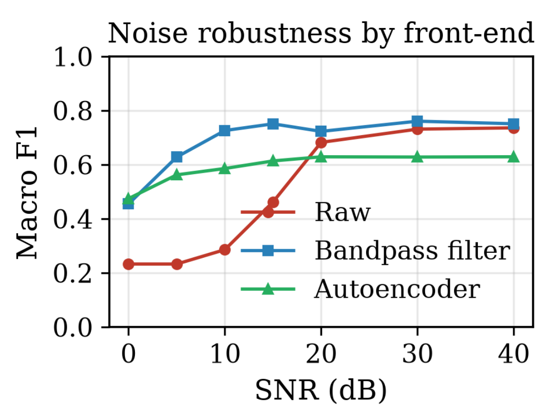

Fig. 3. Macro F1 vs SNR averaged over 5 artifacts (supplementary; see Fig. 5 in the main paper for per-artifact split).

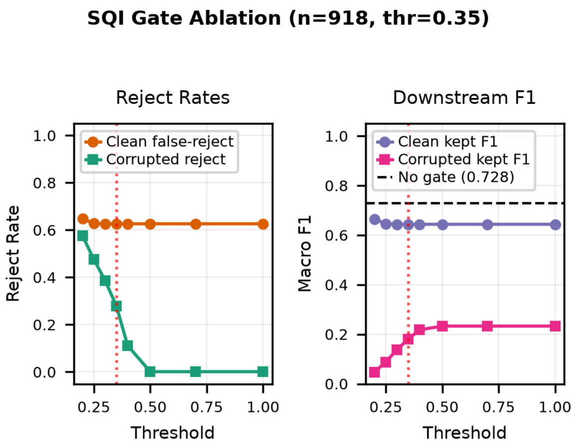

Fig. 4. SQI gate: clean false-reject 62.6% at 0.35, corrupted reject 27.9%, downstream macro 0.727→0.643 — not deployable (Fig. S1 in main text).

TABLE VI
PER-ARTIFACT NOISE ROBUSTNESS (MACRO F1; EXCERPT, SEE JSON FOR FULL 5×7×3).

SNR	Raw	Filter	AE	Artifacts
0–40 dB	0.233–0.736	0.455–0.751	0.475–0.629	avg over 5 (Table III in
Per-artifact extremes (from per_artifact_noise.json): BW: raw 0.71–0.75, filter 0.73–0.79, AE 0.62–0.83; motion: raw wins at 10 dB (0.73 vs 0.68); PLI: AE wins only low-SNR (0.62 at 0 dB); EMG/mixed: filter dominates.				